Aryan Mondal

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Education

SRM Institute of Science and Technology — Chennai, Tamil Nadu

B.Tech Computer Science And Engineering, CGPA: 8.46 — *Sep 2022 - May 2026*

- **Relevant Coursework**: Cryptography and Network Security, Forensics and Incident Response, Security Management, Computer Networks, Cyberwarfare, Mobile and Wireless Security, Hacker Techniques, Tools, and Incident Handling

Experience

Security Intern — Feb 2025 – Apr 2025

Thirumoolar IT Solutions — *Chennai, Tamil Nadu*

- Assisted in performing over 20 vulnerability scans and penetration tests using tools such as OWASP ZAP, Burp Suite, and Nmap, identifying more than 40 security issues and contributing to a 25% improvement in system security posture.
- Reviewed codebases of 3 internal web applications, detecting vulnerabilities like SQL Injection and XSS, and collaborated with developers to achieve a great reduction in security flaws.
- Researched emerging cybersecurity threats and documented findings to enhance the company's internal knowledge base.

Projects

Network Intrusion Detection System — Apr 2025 - Apr 2026

- Built a production-ready network intrusion detection system achieving 99.64% accuracy on the CIC-IDS-2017 dataset (60K samples, 70/15/15 train–validation–test split) with 1,250× fewer parameters than state-of-the-art Transformer models, enabling deployment on Raspberry Pi and edge devices.
- Innovated an adaptive fusion mechanism that learned to weight model components during training, eliminating manual hyperparameter tuning and improving model efficiency by 57% over equal-weight ensembles.
- Reduced false positive rate to 0.08%, critical for minimizing security analyst alert fatigue in production environments.
- Processed 16,000-25,000 network flows per second with 5-8ms batch latency on NVIDIA RTX 3070, demonstrating real-time capability for enterprise network security applications.
- Published findings demonstrating 8.5× parameter efficiency advantage of relational learning over statistical approaches in intrusion detection.

Publications

An Adaptive Feedback Architecture for Stabilizing SNSPDs in TF-QKD Networks using SDQN — Under Review

IEEE Transactions on Quantum Engineering (TQE)

- Deployed a Software-Defined Quantum Network (SDQN) to enable adaptive stabilization of Superconducting Nanowire Single-Photon Detectors (SNSPDs) in the context of Twin-Field Quantum Key Distribution (TF-QKD).
- Achieved 41% Secret Key Rate (SKR) improvement and 4× stability improvement through hot-swappable PID and Bayesian optimization control algorithms, incorporating real-time security and environmental resilience for autonomous, AI-prepared quantum communication networks verified through physics-realistic simulations.

A Robust and Lightweight Authentication Protocol for UAV Swarm Communications using HKDF-based Key Derivation and AEAD — Accepted

IEEE CONECCT 2026

- Designed R-HAAP, replacing chaotic-map-based keystream generation with a formally-verified cryptographic pipeline (ECDH → HKDF → ChaCha20-Poly1305), achieving 150–250 μs per-packet latency on ARM Cortex-M4 — 12–20× faster than software AES-GCM and 20–33× faster than the original Duffing-map protocol.
- Implemented per-packet key derivation using HKDF.expand with UAV ID and monotonic offset counters as context, cryptographically binding packet metadata to the MAC and providing provable resistance against replay and forgery attacks.

- Validated protocol scalability using UavNetSim-v1 (Python/SimPy), demonstrating a stable packet delivery rate exceeding 99.8% and end-to-end latency under 15 ms across swarm sizes of 10–100 UAVs, well within the 100 ms real-time flight control threshold.
- Conducted a multi-layer security analysis covering MITM, replay, cache-timing side-channel, GPS spoofing, and Hardware Trojan threats, demonstrating forward secrecy through ephemeral ECDH session keys that confine any compromise to a single flight session.

Lightweight Hybrid DNN-GNN Architecture for Network Intrusion Detection with Adaptive Late Fusion Apr 2026
ICNWC-2026

- Designed and trained a hybrid deep learning architecture combining a DNN branch (256→128→64→32, ∼68K params) and a Pseudo-GNN branch (64→64→32, ∼8K params) with adaptive late fusion, achieving 99.64% accuracy on the CIC-IDS-2017 dataset across 60,000 network flow samples.
- Implemented a learnable fusion weight α that autonomously converged from 0.689 to 0.389 over 25 training epochs, establishing that relational features carry stronger discriminative power than statistical features for intrusion detection.
- Built an end-to-end pipeline from raw packet capture (Windows raw sockets, 79 CICIDS2017 features) through data preprocessing to real-time inference, classifying over 16,000 flows per second at under 5ms latency with a false positive rate of 0.08%.
- Validated against state-of-the-art architectures including ConvNeXt-Tiny (28M params) and Transformer-based NIDS (100M params), achieving competitive or superior accuracy at 350× and 1,250× fewer parameters respectively.

Quantum Key Distribution for SCADA in Power Grids: Security Architectures and Practical Applications Nov 2025
ICETCE-2025

- Achieved 94.9% classification accuracy in anomaly detection for QKD-enhanced SCADA systems using Support Vector Machines (SVM) and Random Forest models, with SVM attaining 100% recall for anomaly detection.
- Designed and validated models on a dataset of 10,000 random data points, demonstrating robustness of QKD-integrated CPS against cyber threats with consistent performance across precision (0.68–0.70), recall (0.94–1.00), and F1-scores (0.80–0.81).

A Robust Face Expression Recognition System with Music Recommendations Mar 2025
CoaCoNS 2025

- Designed and deployed a real-time facial expression recognition system using a modified VGG-16 architecture, achieving 88% accuracy on the JAFFE dataset (∼213 images across 7 expression classes), leveraging transfer learning and data augmentation techniques.
- Integrated a Haar Cascade-based facial feature detection system and a music recommendation engine, mapping seven expression classes (e.g., happy, sad, surprise) to predefined music playlists.
- Validated through live webcam testing on NVIDIA RTX 3080 hardware, optimized using learning rate scheduling and dropout regularization over 50 training epochs.

Technical Skills

Programming Languages: Python, C, C++, Bash, SQL
Cyber Security & Networking: Cryptography, Digital Forensics, Anomaly Detection, Malware Analysis, Penetration Testing, Web Exploits (SQLi, XSS), Operating Systems, SIEM/SOC Operations, SCADA Security, UAV Security, Quantum Key Distribution (QKD)
Tools & Frameworks: OWASP ZAP, Burp Suite, Nmap, Wireshark, Scapy, PyShark, Apache Kafka, FastAPI, Grafana, TensorFlow, Keras
Other Skills: Risk Management, Compliance & Regulation, Security Awareness, Threat Intelligence, Strategic Planning, Project Management